

if `number` is odd. However, if you'd prefer to overengineer a more “elegant” recursive algorithm, you can implement the following `isOdd()` function in *isOdd.py* (the rest of *isOdd.py* is presented later in this section):

Python

```
def isOdd(number):
    if number == 0:
        # BASE CASE
        return False
    else:
        # RECURSIVE CASE
        return not isOdd(number - 1)
print(isOdd(42))
print(isOdd(99))
--snip--
```

The JavaScript equivalent is in *isOdd.html*:

JavaScript

```
<script type="text/javascript">

function isOdd(number) {
    if (number === 0) {
        // BASE CASE
        return false;
    } else {
        // RECURSIVE CASE
        return !isOdd(number - 1);
    }
}
document.write(isOdd(42) + "<br />");
document.write(isOdd(99) + "<br />");
--snip--
```

We have two base cases for `isOdd()`. When the `number` argument is 0, the function returns `False` to indicate *even*. For simplicity, our implementation of `isOdd()` works for only positive integers. The recursive case returns the opposite of `isOdd(number - 1)`.

You can see why this works with an example: when `isOdd(42)` is called, the function can't determine if 42 is even or odd but does know that the answer